

What is the smallest number of disjoint spanning trees made a graph Hamiltonian

OPG-567

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

We are given a complete simple undirected weighted graph $G_1 = (V, E)$ and its first arbitrary shortest spanning tree $T_1 = (V, E_1)$. We define the next graph $G_2 = (V, E \setminus E_1)$ and find on G_2 the second arbitrary shortest spanning tree $T_2 = (V, E_2)$. We continue similarly by finding $T_3 = (V, E_3)$ on $G_3 = (V, E \setminus \cup_{i=1}^2 E_i)$, etc. Let k be the smallest number of disjoint shortest spanning trees as defined above and let $T^k = (V, \cup_{i=1}^k E_i)$ be the graph obtained as union of all k disjoint trees.

Question 1. What is the smallest number of disjoint spanning trees creates a graph T^k containing a Hamiltonian path.

Question 2. What is the smallest number of disjoint spanning trees creates a graph T^k containing a shortest Hamiltonian path?

Questions 3 and 4. Replace in questions 1 and 2 a shortest spanning tree by a 1-tree. What is the smallest number of disjoint 1-trees creates a Hamiltonian graph? What is the smallest number of disjoint 1-trees creates a graph containing a shortest Hamiltonian cycle?

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Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>